

DATA ANALYTICS PROJECT REPORT

Flipkart E-Commerce

Product, Pricing & Discount Analysis

An end-to-end analysis of 20,000 Flipkart product listings using Microsoft Excel, SQL, Python and Power BI

Prepared by	[Your Name]
Qualification	B.Tech, Computer Science & Engineering
Institution	Pailan College of Management and Technology, MAKAUT (2025)
Role targeted	Data Analyst / Business Analyst / MIS Executive
Tools used	Advanced Excel · MySQL · Python · Power BI
Report date	September 2026

Table of Contents

To populate this list, open the document in Word, press Ctrl+A, then press F9 and choose "Update entire table". Delete this line afterwards.

Table of Contents	2
1. Executive Summary.....	3
2. Project Objectives	3
3. Dataset Overview	4
3.1 Source and scale	4
3.2 Field description	4
4. Tools and Technologies	5
5. Methodology	5
5.1 Stage 1 — Data cleaning in Excel	5
5.2 Stage 2 — SQL analysis.....	5
5.3 Stage 3 — Exploratory data analysis in Python	6
5.4 Stage 4 — Dashboard development	6
6. Analysis and Findings	7
6.1 Category performance	7
6.2 Revenue concentration	8
6.3 Trend over time.....	8
6.4 Pricing and discount structure	9
6.5 Customer ratings	9
6.6 Brand and fulfilment analysis	10
7. Dashboard Walkthrough.....	11
7.1 Power BI dashboard	11
7.2 Excel dashboard	12
8. Key Insights	14
9. Business Recommendations	15
9.1 Protect and extend the Jewellery franchise	15
9.2 Restructure the discount policy	15
9.3 Fix the ratings coverage gap	15
9.4 Intervene on low-rated, high-value categories	15
9.5 Expand Flipkart Advantage to premium inventory	15
9.6 Consolidate the brand base	15
10. Limitations.....	16
11. Conclusion	16
Appendix A — SQL Query Reference.....	17
Appendix B — Project Deliverables	17

1. Executive Summary

This report presents an end-to-end analysis of a Flipkart e-commerce product catalogue containing 20,000 raw listings. The objective was to convert an unstructured, partially incomplete web-scraped dataset into a reliable analytical base, and then to answer commercial questions about category performance, pricing structure, discount strategy and customer satisfaction.

The raw file was cleaned in Microsoft Excel using Power Query, reducing it to 13,754 analysis-ready records across 13 standardised fields. The cleaned data was then loaded into MySQL for structured querying, explored in Python using Pandas, NumPy, Matplotlib and Seaborn, and finally visualised in an interactive Power BI dashboard supported by a secondary dashboard built in Excel.

Headline results from the cleaned dataset:

- **Jewellery is the dominant category** — 3,522 listings, the largest of any category and 25.6% of the catalogue.
- **Revenue is heavily concentrated** — **Jewellery alone accounts for** ₹19.58 million of ₹33.69 million total discounted-price value, or 58.1% of the entire catalogue value, despite being a single category.
- **Discounting is aggressive** — **the average product is sold at** 42.8% off retail price, but the spread is wide: 1,234 products carry no discount at all while 1,624 are discounted by more than 70%.
- **Ratings vary sharply by category** — Beauty & Personal Care (4.65), Footwear (4.32) and Home Decor (4.31) lead on average rating, while Furniture (2.83) and Home Improvement (2.25) lag well behind.
- **Rating coverage is very poor** — only 904 of 13,754 products (6.6%) carry a customer rating, which is itself a significant operational finding.

Bottom line

The catalogue is commercially lopsided. A small number of categories generate almost all of the value, deep discounting is applied without an obvious link to customer satisfaction, and the review programme covers barely one product in fifteen. Each of these is addressable, and Section 9 sets out the recommended actions.

2. Project Objectives

The project was designed to demonstrate the complete analytics workflow — from raw data to business recommendation — rather than any single tool. The specific objectives were:

1. Clean and standardise a large, messy, web-scraped e-commerce dataset into a trustworthy analytical table.
2. Quantify category performance by order volume, revenue contribution and average selling price.
3. Analyse the pricing and discount structure and identify where discounting is excessive or inconsistent.
4. Assess customer satisfaction by category and measure the coverage of the ratings programme.
5. Identify the leading brands and the highest-value products in the catalogue.

- Deliver the findings through an interactive dashboard and translate them into actionable business recommendations.

3. Dataset Overview

The source file, `flipkart_com-ecommerce_sample.csv`, is a publicly available web-scrape of the Flipkart marketplace. It contains 20,000 rows and 15 columns of product-level information captured between December 2015 and June 2016.

3.1 Source and scale

Attribute	Detail
Source file	<code>flipkart_com-ecommerce_sample.csv</code>
Raw records	20,000 rows × 15 columns
Cleaned records	13,754 rows × 13 columns
Records removed	6,246 (31.2%) — duplicates, null prices, unusable category strings
Period covered	1 December 2015 to 28 June 2016
Distinct categories	31
Distinct brands	3,394
Currency	Indian Rupee (₹)

Table 1 — Dataset profile before and after cleaning

3.2 Field description

The cleaned table, `cleanData.csv`, uses the following schema:

Field	Type	Description
<code>uniq_id</code>	Text	Unique identifier for each listing; used as the order-count key
<code>Purches_Date</code>	Date	Purchase date, derived by splitting the original <code>crawl_timestamp</code>
<code>Purches_Time</code>	Time	Purchase time, derived from the same source field
<code>product_url</code>	Text	Listing URL on <code>flipkart.com</code>
<code>product_name</code>	Text	Product title as displayed on the listing
<code>product_category</code>	Text	Top-level category extracted from <code>product_category_tree</code>
<code>pid</code>	Text	Flipkart internal product ID
<code>retail_price</code>	Integer	Original list price in ₹ (MRP)
<code>discounted_price</code>	Integer	Actual selling price in ₹
<code>is_FK_Advantage_product</code>	Boolean	Flag for Flipkart Advantage fulfilment
<code>product_rating</code>	Text	Product-level rating, 1 to 5; "No rating available" where absent
<code>overall_rating</code>	Text	Aggregate seller/product rating
<code>brand</code>	Text	Brand name

Table 2 — Data dictionary for `cleanData.csv`

4. Tools and Technologies

Tool	Role in the project	Specific techniques applied
Microsoft Excel	Data cleaning and first-pass analysis	Power Query, Text-to-Columns, Remove Duplicates, XLOOKUP, INDEX-MATCH, Pivot Tables, Slicers, conditional formatting, Excel dashboard
MySQL	Structured querying and validation	GROUP BY aggregation, CASE expressions, scalar subqueries, RANK() window function partitioned by category
Python	Exploratory data analysis	Pandas for aggregation and null handling, NumPy for numeric operations, Matplotlib and Seaborn for distribution and relationship plots
Power BI	Interactive reporting	Data model, DAX measures, KPI cards, category and brand visuals, slicers for date and category drill-down
Microsoft Word	Documentation	This report

Table 3 — Technology stack

5. Methodology

The project followed a four-stage pipeline. Each stage produced a deliverable that fed into the next, so the analysis is reproducible end to end.

5.1 Stage 1 — Data cleaning in Excel

The raw extract was unusable in its delivered form. The following treatments were applied, producing cleanData.csv:

- **Timestamp separation** — crawl_timestamp was split into two usable fields, Purches_Date and Purches_Time, so that time-series analysis became possible.
- **Category extraction** — product_category_tree stored a full breadcrumb path in a single cell. The top-level category was extracted into a clean product_category field, collapsing free-text paths into 31 usable categories.
- **Column removal** — image, description and product_specifications were dropped. These are long free-text and URL fields of no analytical value in this scope, and their removal reduced file size by roughly 90%.
- **Null price handling** — rows with missing or zero retail_price or discounted_price were removed, since every pricing and discount metric depends on them.
- **Duplicate removal** — exact duplicate listings were removed using Remove Duplicates on uniq_id.
- **Type enforcement** — price fields were cast to whole-number integers and the Advantage flag was cast to a Boolean.

Outcome: 13,754 clean records from 20,000 raw rows, a retention rate of 68.8%. All 13 fields are complete apart from three missing brand values (0.02%).

5.2 Stage 2 — SQL analysis

The cleaned file was imported into a MySQL database, FlipcartECommers, and queried to answer the core business questions. Nine queries were written, covering order counts by category, discounted value by date, average rating by category, high-value brand filtering, a CASE-based price banding (Low under ₹1,000, Moderate ₹1,000–₹4,000, High above ₹4,000), subquery comparisons against the catalogue average price, and a RANK() window function returning the most expensive products within each category. The full query set is reproduced in Appendix A.

5.3 Stage 3 — Exploratory data analysis in Python

The dataset was loaded into a Jupyter notebook (EDA.ipynb) with Pandas. Structure and completeness were profiled using `.info()`, `.describe()`, `.isna().sum()` and `.duplicated().sum()`. Seaborn and Matplotlib were then used to visualise revenue by category, brand frequency, the relationship between rating and price, the effect of the Flipkart Advantage flag on selling price, and category activity over time.

5.4 Stage 4 — Dashboard development

Two dashboards were produced. The Excel dashboard in Analysis.xlsx is built on pivot tables with slicers, giving a lightweight offline view of category volume, price totals and average ratings. The Power BI dashboard, dashboard.pbix, is the primary deliverable: it presents KPI cards for total orders, total revenue and average discount, alongside category and brand breakdowns, a monthly trend line, and slicers allowing the user to filter by date range and category.

6. Analysis and Findings

6.1 Category performance

Order volume is highly concentrated. Jewellery (3,522 listings) and Clothing (3,135) together account for 48.4% of the entire catalogue, while the remaining 29 categories share the balance. Beyond the top six, volumes fall away sharply — categories such as Watches, Eyewear and Wearable Smart Devices carry fewer than 50 listings each.

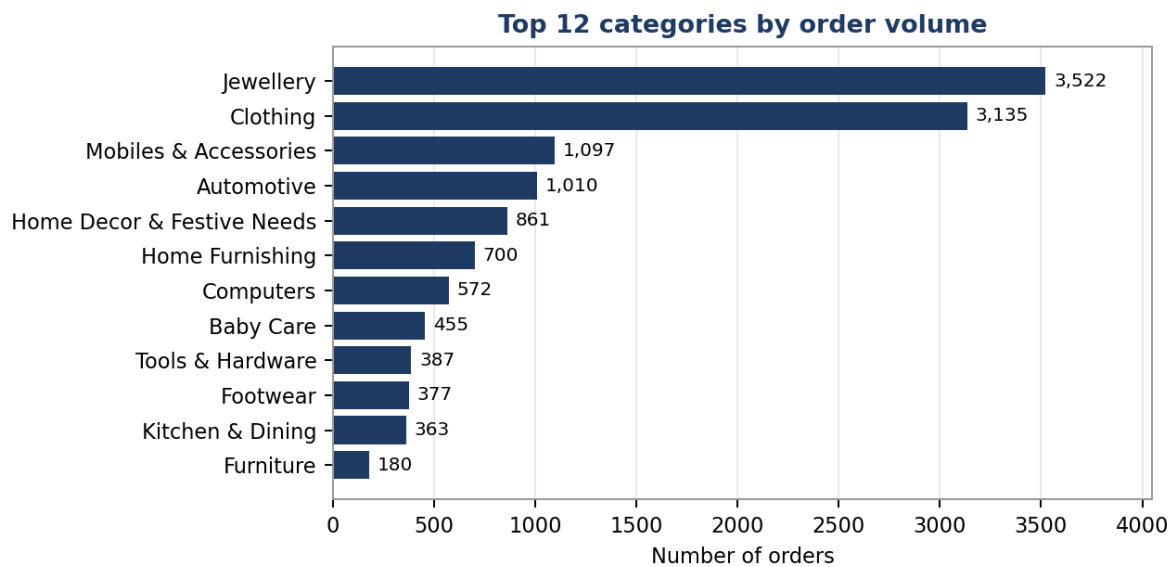


Figure 1 — Top 12 categories by order volume

The following table summarises performance across the twelve largest categories. Revenue is measured as the sum of discounted_price, the price at which items actually sell.

Category	Orders	Revenue (₹M)	Avg retail (₹)	Avg disc.	Avg rating
Jewellery	3,522	19.58	7,663	40.9%	4.09
Clothing	3,135	2.27	1,473	50.0%	3.74
Mobiles & Accessories	1,097	1.05	1,810	51.0%	3.76
Automotive	1,010	1.20	2,903	54.7%	3.67
Home Decor & Festive Needs	861	1.65	2,867	36.2%	4.31
Home Furnishing	700	0.86	1,942	37.2%	3.95
Computers	572	1.19	3,238	43.5%	3.88
Baby Care	455	0.40	1,453	35.0%	4.07
Tools & Hardware	387	0.34	1,320	29.3%	3.38
Footwear	377	0.29	1,265	36.0%	4.33
Kitchen & Dining	363	0.30	1,187	33.3%	4.11
Furniture	180	3.08	23,263	29.7%	2.83

Table 4 — Performance summary for the twelve largest categories

6.2 Revenue concentration

Volume and value do not move together. Jewellery contributes ₹19.58 million, or 58.1% of the ₹33.69 million total catalogue value, because it combines high volume with a high average retail price of ₹7,663. Furniture illustrates the opposite pattern: just 180 listings, yet ₹3.08 million in value — second overall — on the strength of an average retail price of ₹23,263. Clothing, despite being the second-largest category by volume, contributes only ₹2.27 million because its average price is low and its discount is deep.

The top five categories together account for 82.5% of catalogue value, a textbook Pareto distribution.

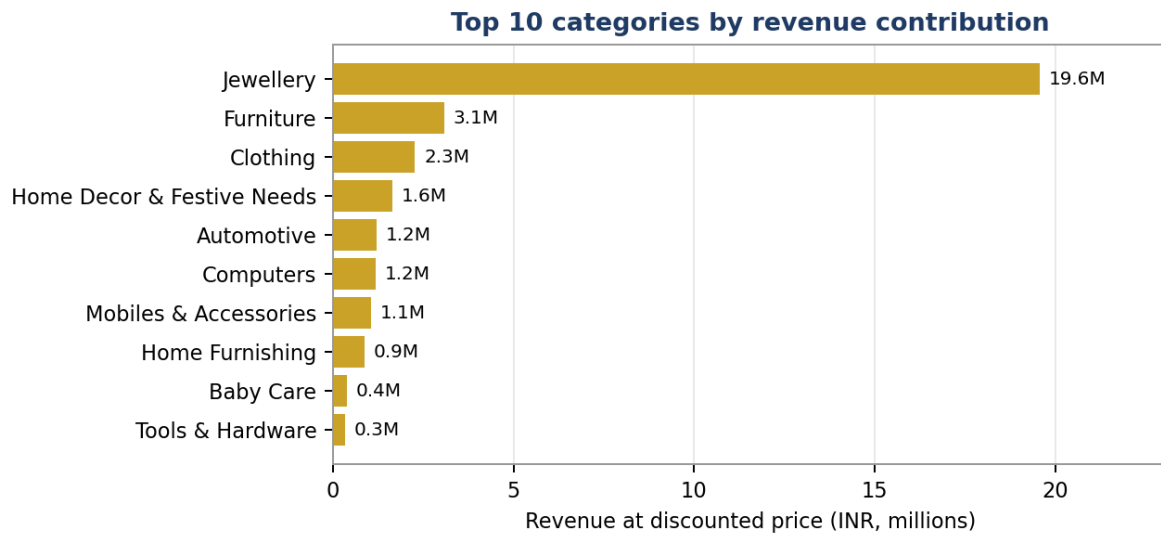


Figure 2 — Top 10 categories by revenue contribution

6.3 Trend over time

Activity is not evenly distributed across the seven-month window. December 2015 carries the highest listing count at 6,054, consistent with festive and year-end trading, while January 2016 produces the highest value at ₹20.0 million on only 3,201 listings — a month dominated by high-ticket items. February 2016 collapses to 244 listings, and activity then recovers to a lower, flatter level through to June.

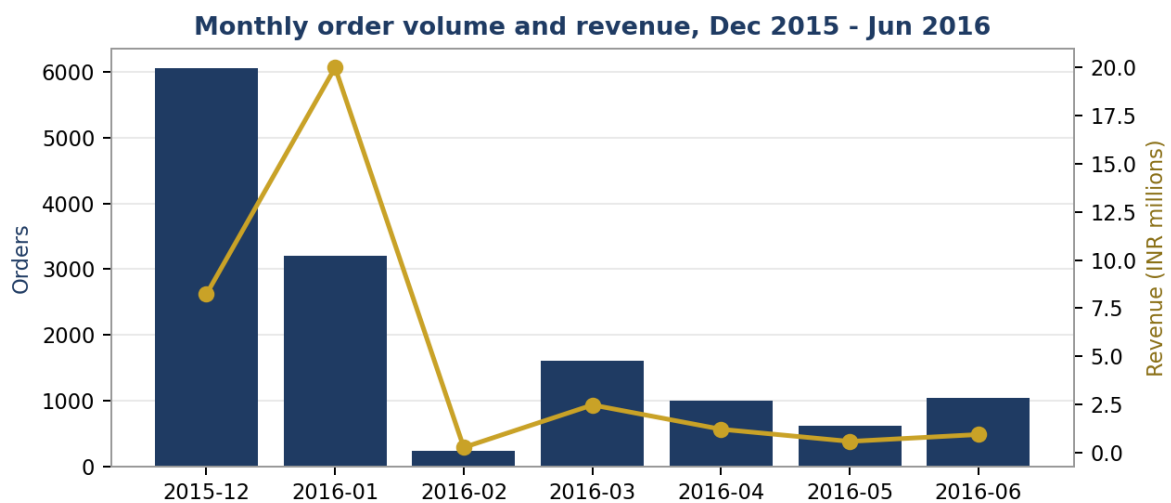


Figure 3 — Monthly order volume and revenue, December 2015 to June 2016

The February trough is almost certainly an artefact of the underlying web-scrape rather than a genuine collapse in trading, and is noted as a limitation in Section 10.

6.4 Pricing and discount structure

Across the catalogue, the average retail price is ₹3,684 and the average selling price ₹2,449, giving a mean discount of 42.8%. The median values are considerably lower — ₹1,100 retail and ₹549 selling — which confirms that a small number of very expensive products are pulling the average upward. The most expensive item in the catalogue is a Durian Leather 2 Seater Sofa at ₹250,500 retail.

The distribution of discount depth is strikingly bimodal:

- 1,234 products (9.0%) carry no discount whatsoever and sell at full MRP.
- 1,624 products (11.8%) are discounted by more than 70%, which for most retail categories is below a sustainable margin.
- The middle 50% of products sit between 25.1% and 60.8% discount — itself a very wide band for a single marketplace.

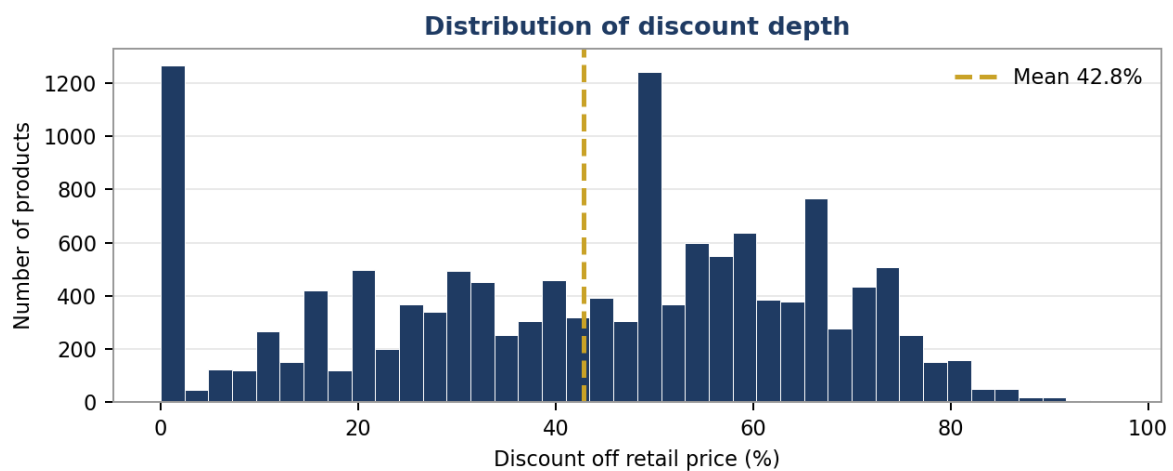


Figure 4 — Distribution of discount depth across 13,754 products

Discounting also differs sharply by category. Watches (57.4%), Automotive (54.7%), Mobiles & Accessories (51.0%) and Clothing (50.0%) are discounted at roughly twice the rate of Beauty & Personal Care (27.5%), Toys (28.2%) and Pens & Stationery (28.0%).

6.5 Customer ratings

Only 904 of 13,754 products carry a usable numeric rating — a coverage rate of 6.6%. The remaining 93.4% are recorded as "No rating available". Where ratings do exist, the catalogue average is 3.85 out of 5, and the distribution is polarised: the 25th percentile sits at 3.0 and the 75th at 5.0, indicating that products tend to be rated either very well or poorly rather than clustering in the middle.

By category, Beauty & Personal Care leads at 4.65, followed by Footwear (4.32), Home Decor & Festive Needs (4.31) and Kitchen & Dining (4.11). At the other end, Furniture averages 2.83 and Home Improvement 2.25. Two categories with meaningful volume — Bags, Wallets & Belts (151 listings) and Watches (48) — carry no numeric ratings at all.

Plotting discount depth against average rating shows no positive relationship between the two. The most heavily discounted categories — Automotive, Mobiles & Accessories, Clothing — all sit below the catalogue average rating, while the best-rated category, Beauty & Personal Care, is the least discounted of all.

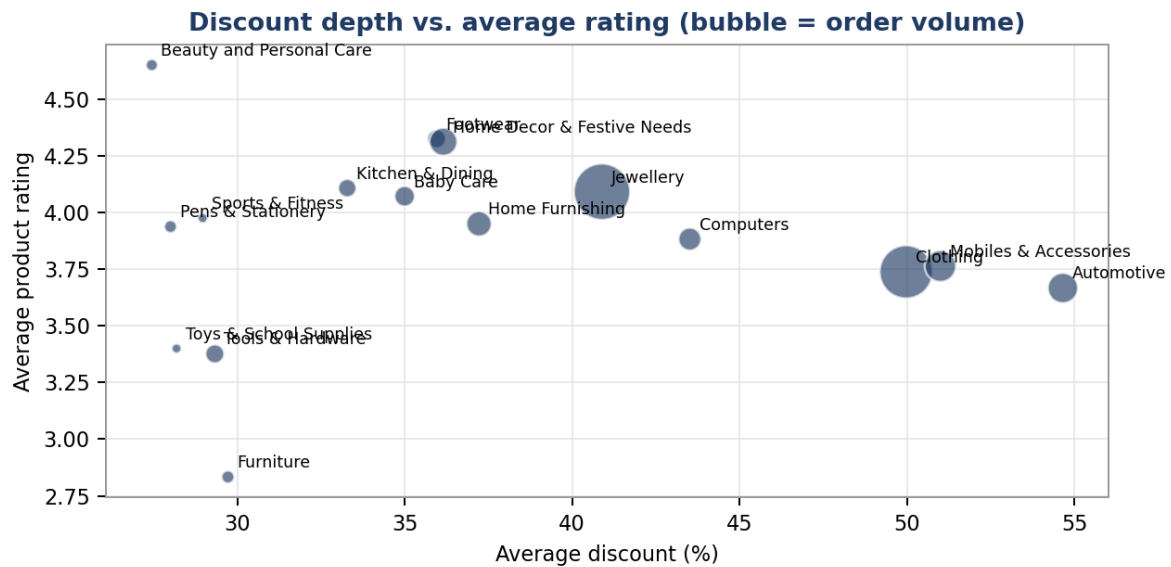


Figure 5 — Discount depth versus average rating; bubble size represents order volume

Key observation

Deep discounting is not buying customer satisfaction. Across the twelve largest categories, the ones with the heaviest markdowns are consistently among the lowest rated. This suggests discount is being used to move weak inventory rather than to reward or attract customers to strong products.

6.6 Brand and fulfilment analysis

The catalogue is extremely fragmented, with 3,394 distinct brands across 13,754 listings — an average of just four listings per brand. The largest brand, Allure Auto, accounts for 468 listings (3.4%), followed by Voylla (299), Slim (284), TheLostPuppy (229) and Karatcraft (211). No brand holds more than 3.5% of the catalogue, so brand-level negotiating leverage is minimal.

Flipkart Advantage fulfilment is applied to only 334 listings, or 2.4% of the catalogue. Advantage products have a lower average retail price (₹2,289 against ₹3,718) and a shallower average discount (38.2% against 42.9%), suggesting the programme is currently being applied to mid-priced, faster-moving goods rather than to premium inventory.

7. Dashboard Walkthrough

Two dashboards accompany this report. The Power BI file, dashboard.pbix, is the primary interactive deliverable; the Excel dashboard in Analysis.xlsx provides a lightweight offline equivalent built entirely on pivot tables and slicers.

7.1 Power BI dashboard

The report is built on a single page with a branded header, a slicer panel down the left, two KPI cards along the top, and a grid of category, brand, time and rating visuals beneath. All visuals are cross-filtered, so selecting a category, quarter or seller in the slicer panel updates the entire page.

Visual	Type	Fields and purpose
Count of Product	KPI card	Product count — total listings in the current filter context
Total Revenue	KPI card	SUM(discounted_price) — catalogue value at actual selling price
Quarter slicer	Slicer	Purches_Date date hierarchy at quarter level
Category slicer	Slicer	product_category — filters the full page to one or more categories
Seller slicer	Slicer	brand — allows single-seller drill-down
Revenue by category	Bar chart	SUM(discounted_price) by product_category — identifies value concentration
Category share	Donut chart	Count of products by product_category — shows assortment mix
Revenue trend	Line chart	SUM(discounted_price) by quarter — seasonality and trend
Orders by time of day	Line chart	Count of uniq_id by Purches_Time — intraday activity pattern
Rating distribution	Pie chart	Count of uniq_id by product_rating — spread of customer scores
Brand contribution	Treemap	Count of uniq_id by brand — surfaces the largest sellers
Orders by category	Clustered bar	Count of uniq_id by product_category — volume ranking

Table 5 — Visual inventory of the Power BI dashboard

Figure 6 — Power BI dashboard, default view (all filters cleared)

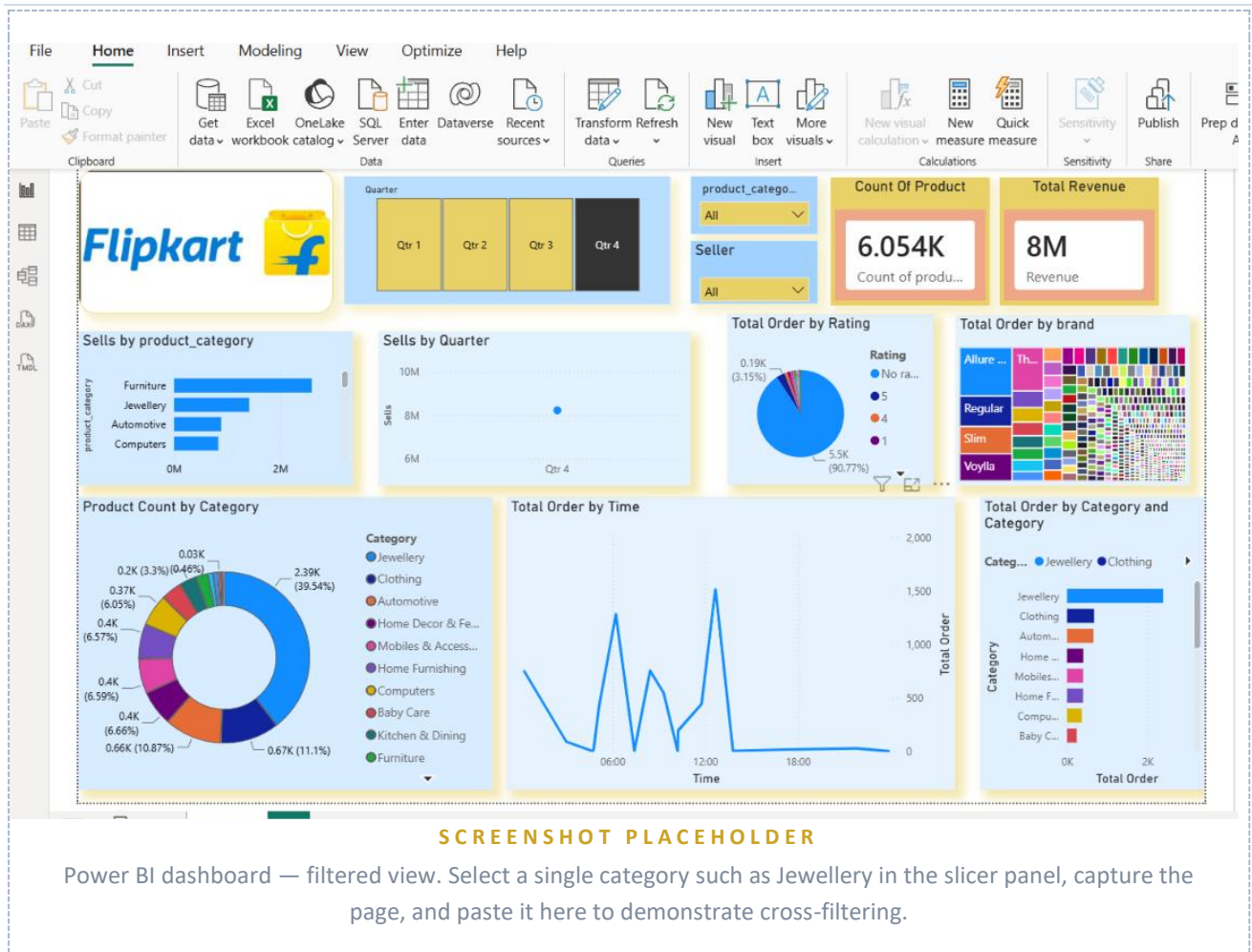


Figure 7 — Power BI dashboard filtered to a single category, showing cross-filter behaviour

7.2 Excel dashboard

The Excel dashboard, on the "Excel Dashboard" sheet of Analysis.xlsx, is driven by the pivot tables on the adjacent sheet. It carries a title banner, two slicers for category and date, a total retail price by category chart, and an average rating by category chart. A companion sheet records the key insights and business recommendations directly alongside the visuals.

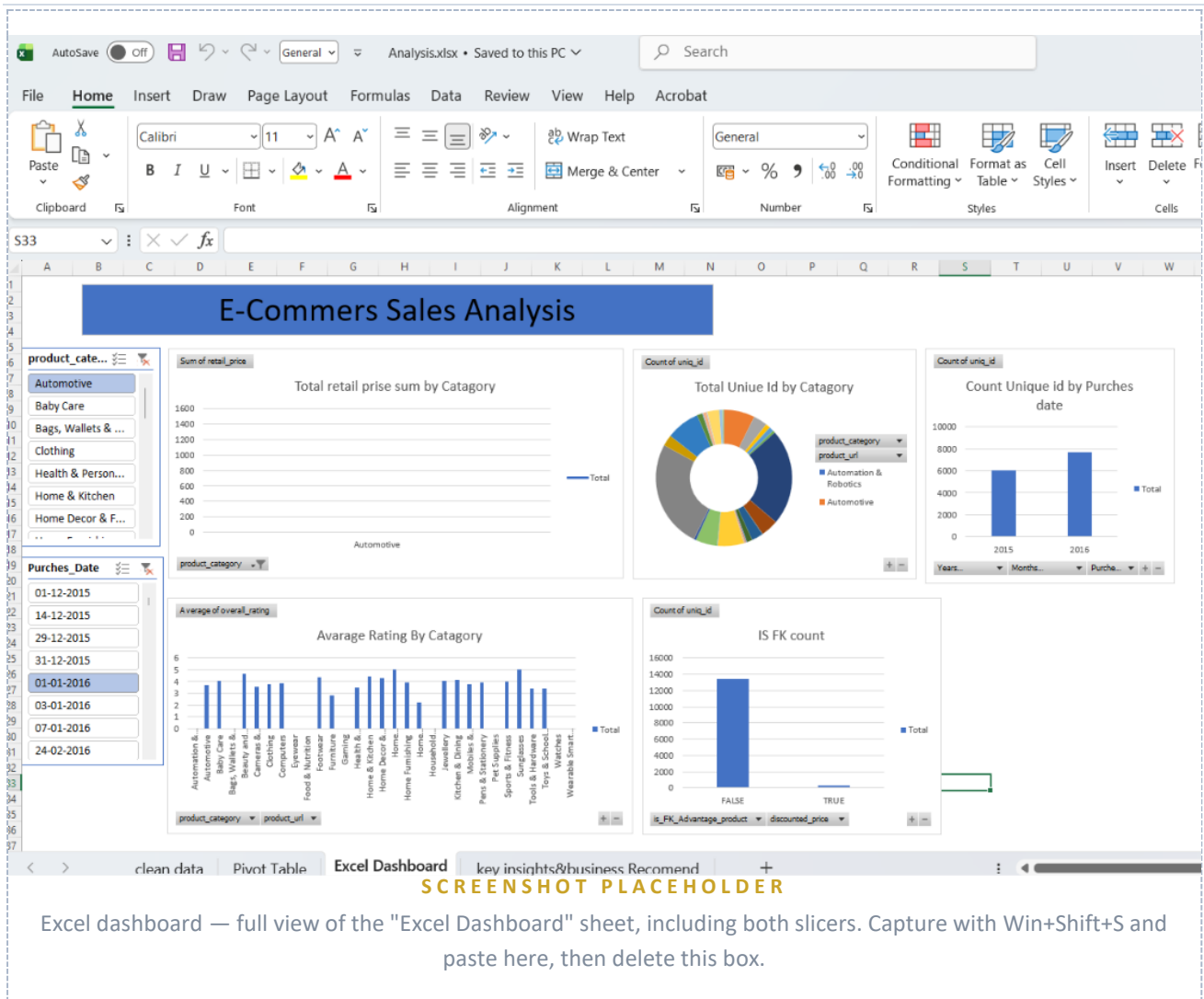


Figure 8 — Excel dashboard built on pivot tables with interactive slicers

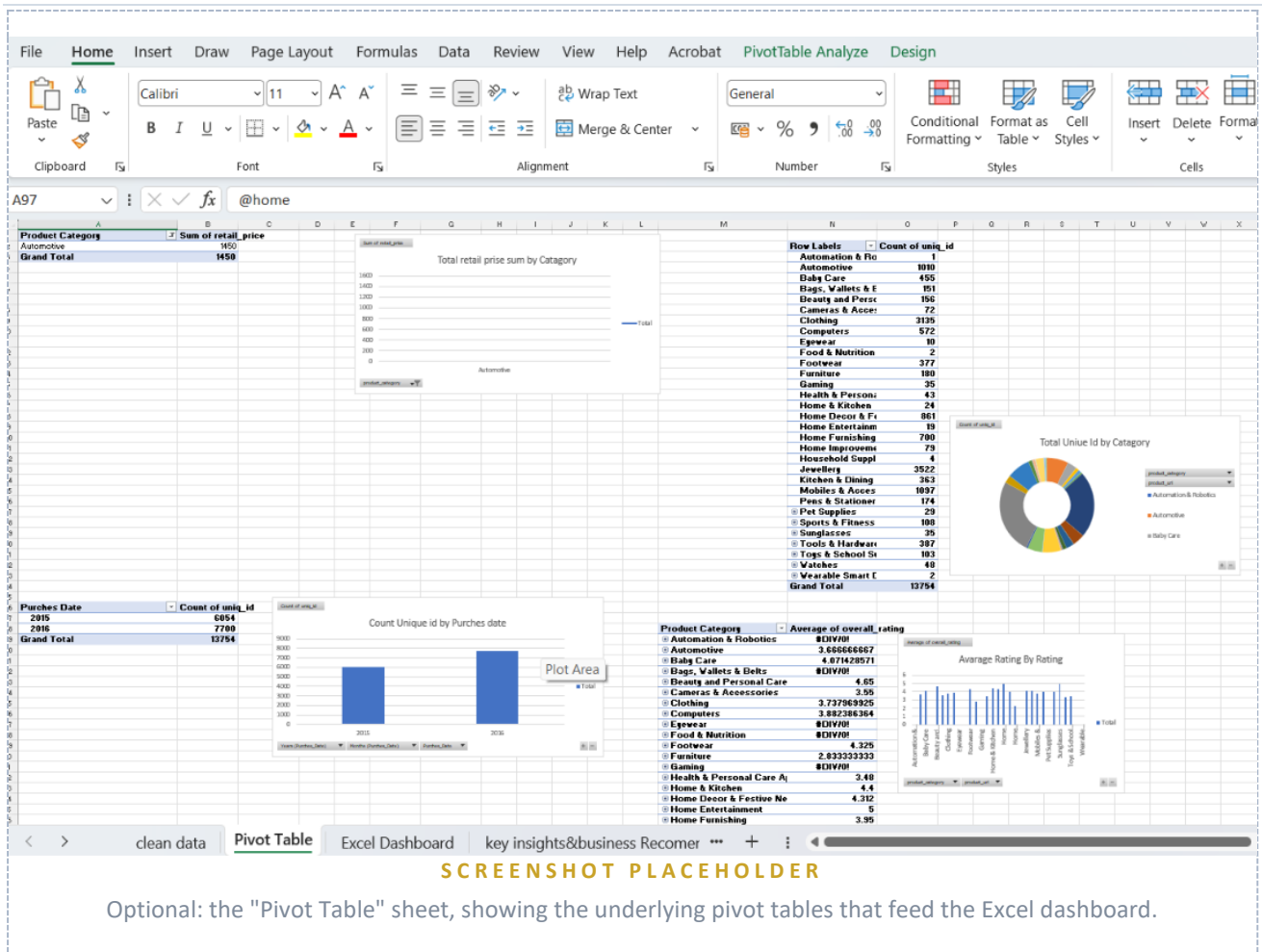


Figure 9 — Supporting pivot tables

8. Key Insights

- Jewellery is the single most important category in the catalogue, leading on both order volume (3,522 listings) and revenue (₹19.58 million, 58.1% of total value).
- Value is concentrated in five categories, which together generate 82.5% of catalogue value; the remaining 26 categories contribute less than one-fifth between them.
- High volume does not imply high value. Clothing ranks second on volume but only third on revenue, while Furniture ranks twelfth on volume and second on revenue.
- Discounting is inconsistent and unusually deep. The mean discount is 42.8%, with 11.8% of the catalogue marked down by more than 70% and 9.0% not discounted at all.
- There is no observable link between discount depth and customer rating; the most discounted categories are among the lowest rated.
- The ratings programme covers only 6.6% of the catalogue, making satisfaction measurement unreliable for the great majority of products.
- Furniture (2.83) and Home Improvement (2.25) average ratings well below the catalogue mean of 3.85, and Furniture is simultaneously a high-value category — a combination that carries real returns and reputational risk.
- Brand concentration is negligible: no single brand exceeds 3.5% of listings across 3,394 brands.

15. Flipkart Advantage fulfilment reaches only 2.4% of listings and is skewed toward lower-priced products.

9. Business Recommendations

The following recommendations follow directly from the findings above and are ordered by expected commercial impact.

9.1 Protect and extend the Jewellery franchise

- Jewellery drives 58% of catalogue value on a moderate 40.9% discount and a healthy 4.09 rating, making it the strongest-performing major category. Increase assortment depth, secure priority inventory from Voylla and Karatcraft, and give the category prominent placement in seasonal campaigns.

9.2 Restructure the discount policy

- Introduce a category-level discount ceiling. The 1,624 products discounted above 70% should be reviewed individually; where the markdown does not clear stock or lift ratings, it is destroying margin without purchase.
- Reduce blanket discounting in Automotive (54.7%) and Mobiles & Accessories (51.0%), where deep markdowns coincide with below-average ratings. Redirect that margin into fulfilment quality and merchandising.
- Test selective discounting on the 1,234 products currently at full MRP to establish whether they are genuinely price-inelastic or simply under-promoted.

9.3 Fix the ratings coverage gap

- A 6.6% rating coverage rate leaves 93% of the catalogue commercially blind. Launch post-delivery review prompts, incentivised reviews, and a verified-buyer badge, with an initial target of 30% coverage within two quarters. Prioritise Bags, Wallets & Belts and Watches, which currently have no ratings at all.

9.4 Intervene on low-rated, high-value categories

- Furniture combines the second-highest revenue contribution with the catalogue's weakest ratings (2.83). Audit the top Furniture sellers for delivery damage, assembly and description accuracy, and delist persistent offenders. Apply the same review to Home Improvement (2.25) and Tools & Hardware (3.38).

9.5 Expand Flipkart Advantage to premium inventory

- Advantage currently covers 2.4% of listings, concentrated at lower price points. Extending guaranteed fulfilment to high-value Jewellery, Furniture and Computers listings would address the exact categories where delivery quality most affects satisfaction and return rates.

9.6 Consolidate the brand base

- With 3,394 brands averaging four listings each, catalogue management costs are high and buying leverage is low. Identify the top 100 brands by revenue and rating, deepen those relationships, and prune long-tail brands that are both low-volume and low-rated.

10. Limitations

- The dataset is a product catalogue scrape, not a transaction log. "Orders" in this report means listings, so revenue figures represent catalogue value rather than confirmed sales.
- The 6,246 records removed during cleaning (31.2% of the raw file) may not be missing at random; if expensive or poorly rated items were disproportionately incomplete, some figures will be biased.
- Rating coverage of 6.6% means category rating averages rest on small samples. Watches and Bags, Wallets & Belts have no numeric ratings at all and are excluded from rating comparisons.
- The sharp drop to 244 records in February 2016 is more consistent with a gap in the data collection than with genuine trading behaviour, so month-on-month trend conclusions should be treated with caution.
- The data covers December 2015 to June 2016. Pricing and category dynamics in Indian e-commerce have changed substantially since; the findings demonstrate analytical method rather than current market conditions.
- No cost or margin data is available, so discount analysis measures markdown off MRP, not profitability.

11. Conclusion

This project took a 20,000-row, incomplete, web-scraped product catalogue and turned it into a 13,754-record analytical asset supported by SQL queries, Python exploratory analysis and two working dashboards. The resulting analysis shows a marketplace with a clear commercial core — Jewellery and a small group of high-value categories carrying most of the value — sitting alongside three fixable weaknesses: undisciplined discounting, near-absent ratings coverage, and quality problems in high-value categories.

Beyond the findings themselves, the project demonstrates the complete analyst workflow: defining business questions, cleaning and validating data, querying it with SQL, exploring it in Python, visualising it for a non-technical audience, and stating recommendations that a commercial team could act on.

Appendix A — SQL Query Reference

The following queries were executed against the FlipkartECommers MySQL database. Each is annotated with the business question it answers.

Business question	SQL technique used
How many listings does each category contain?	COUNT(uniq_id) with GROUP BY product_category
What is the total discounted value by date?	SUM(discounted_price) with GROUP BY Purches_date
What is the average rating by category?	AVG(product_rating) with GROUP BY product_category
Which brands carry premium-priced products?	WHERE retail_price > 1500 with ORDER BY retail_price DESC
How do products band by price?	CASE WHEN expression: Low (<₹1,000), Moderate (₹1,000–₹4,000), High (>₹4,000)
Which products price above the catalogue average?	Scalar subquery: WHERE retail_price > (SELECT AVG(retail_price) ...)
What are the most expensive products in each category?	RANK() OVER (PARTITION BY product_category ORDER BY retail_price DESC)

Table 6 — SQL query inventory

Appendix B — Project Deliverables

File	Contents
flipkart_com-ecommerce_sample.csv	Raw source dataset, 20,000 rows × 15 columns
cleanData.csv	Cleaned analytical dataset, 13,754 rows × 13 columns
Analysis.xlsx	Cleaned data, pivot tables, Excel dashboard, and an insights and recommendations sheet
queries.sql	Nine annotated MySQL queries covering aggregation, banding, subqueries and window functions
EDA.ipynb	Jupyter notebook: profiling, null and duplicate handling, and Seaborn visualisations
dashboard.pbix	Interactive Power BI dashboard with KPI cards, category and brand visuals, and date slicers
Project_Report.docx	This report

Table 7 — Files submitted with this project